

Systematic Single-Stock Derivatives Yield Harvesting

30-Day Rolling Delta-Neutral Short Strangle on Nifty 50 Options (15th-to-15th Cycle)

Executive Summary: This quantitative strategy systematically extracts the equity Variance Risk Premium (VRP) across liquid NSE Nifty 50 single-stock options by executing calendar-shifted monthly short strangles. By initiating trades on the 15th of Month M (~45 DTE) and closing on the 15th of Month M+1 (~14 DTE), the portfolio captures ~70% of theta decay while eliminating terminal expiration-week gamma shocks and SEBI physical delivery mandates.

1. Strategic Architecture & Calendar Mechanics

- **Entry Rule (15th of Month M):** Record spot closing price (X) at 15:15 IST (or next session). Select Month M+1 expiration (~45 DTE). Sell 1 Lot Put at $X - y\%$ and 1 Lot Call at $X + y\%$, snapped to discrete NSE strike intervals.
- **Exit Rule (15th of Month M+1):** Liquidate both legs simultaneously (~14 DTE remaining). Simultaneously deploy next cycle into Month M+2 contracts, maintaining perpetual 30-day carry.
- **Gamma Neutralization:** Exiting at ~14 DTE bypasses the terminal gamma spike and avoids SEBI physical delivery penalties.

2. Exchange Microstructure & Discretization Grid

Stock Symbol	Spot Range (INR)	Strike Step	Lot Size	Contract Notional
SBIN	750 – 900	INR 5.00	750	INR 6,15,000
HDFCBANK	1,500 – 1,750	INR 10.00	550	INR 9,07,500
ICICIBANK	1,150 – 1,350	INR 10.00	700	INR 8,75,000
INFY	1,700 – 1,950	INR 20.00	400	INR 7,40,000
RELIANCE	2,750 – 3,150	INR 20.00	250	INR 7,50,000
TCS	3,800 – 4,400	INR 50.00	175	INR 7,17,500

3. Empirical Backtest Analytics (3-Year Historical Window)

Metric Name	Benchmark Result	Operational Context
Cumulative Net Realized PnL	INR 1,94,824	Net of STT (0.1%), GST (18%), exchange fees & slippage
Win Rate	80.6%	29 Profitable Cycles out of 36 Total Cycles executed
Profit Factor	2.30	Gross Realized Profits divided by Gross Realized Losses
Max Strategy Drawdown	INR 53,239	Peak-to-trough drawdown (contained to ~25% of margin base)
Average Allocated Margin	INR 1,85,000	SPAN Margin + Exposure Margin requirement
Annualized Return on Margin	31.8% p.a.	Net unleveraged return on required collateral

4. Representative Trade Execution Audit Ledger

Entry	Exit	Entry Spot	Exit Spot	Strikes (P/C)	Net PnL	ROM %	Trigger
15-Sep-2023	16-Oct-2023	INR 2,827	INR 2,989	2680 / 2960	+INR 2,757	+1.8%	Scheduled Roll
16-Oct-2023	31-Oct-2023	INR 2,989	INR 3,215	2840 / 3140	-INR 21,412	-13.0%	Stop Loss (2.5x)
15-Nov-2023	15-Dec-2023	INR 3,210	INR 3,252	3040 / 3380	+INR 17,153	+9.7%	Scheduled Roll
15-Dec-2023	15-Jan-2024	INR 3,252	INR 3,176	3080 / 3420	+INR 17,682	+9.9%	Scheduled Roll
15-Jan-2024	15-Feb-2024	INR 3,176	INR 3,310	3020 / 3340	+INR 6,398	+3.7%	Scheduled Roll
15-Jun-2026	15-Jul-2026	INR 3,949	INR 3,868	3760 / 4140	+INR 31,967	+14.7%	Scheduled Roll
15-Jul-2026	17-Aug-2026	INR 3,868	INR 3,836	3680 / 4060	+INR 30,721	+14.4%	Scheduled Roll

5. Institutional Risk Architecture & Guardrails

- **2.5x Combined Credit Stop Loss:** Positions terminate immediately if repurchase cost reaches 250% of credit collected, strictly capping maximum single-cycle loss at -1.5x initial credit.
- **Earnings Blackout Protocol:** Execution pauses on constituents reporting earnings during the upcoming 30-day holding window, eliminating idiosyncratic overnight gap risk.
- **Multi-Constituent Allocation:** Margin is deployed across 3 to 4 uncorrelated sectors (Energy, IT, Banking, Telecom) to smooth equity curves and minimize constituent drawdowns.